Fiscal Year:	FY 2023	**Task Last Updated:**	FY 10/26/2022
PI Name:	Lee, Stuart M.C. Ph.D.		
Project Title:	Effect of Sex on Orthostatic Intolerance and Cardiovascular Responses during Lunar Descent and Ascent		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/17/2021	**End Date:**	12/15/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/15/2024 per C. Ribeiro/JSC (Ed., 3/21/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lytle, Jason Ph.D. (KBR/NASA Johnson Space Center) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Macias, Brandon Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
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Task Description:	Cabin design configurations for the Human Landing System (HLS) have not yet been finalized and may include standing with head-to-foot accelerations during the lunar descent and ascent. None of the Apollo astronauts reported symptoms of orthostatic intolerance while standing in the Lunar Module during sustained gravity levels (G-levels) ranging from 0.4- to 0.6 G over ~9 minutes during lunar descent and ascent, but almost all experienced elevated heart rates, likely resulting from the combination of some amount of cardiovascular deconditioning known to occur with weightlessness, the head-to-foot acceleration, and the psychological stressors during these phases of flight. The G-levels to which the Artemis astronauts will be exposed have not been defined, but it is likely that the G-levels will be greater and/or the exposures will be longer (up to 20 minutes) than those on Apollo missions, and thus may represent a greater challenge to the spaceflight-adapted cardiovascular system. All the Apollo astronauts were male, but the proportion of women in the astronaut corps has increased and the first Artemis mission to the surface of the moon will include at least one woman. Because women are more susceptible to post-spaceflight orthostatic intolerance when re-exposed to Earth's gravity, there is a need for orthostatic intolerance data in both sexes across varying gravity levels to inform NASA standards to safely protect all astronauts during lunar descent and ascent. Further, there are limited data, particularly with regard to women, as to the efficacy of the lower body compression garments to protect against orthostatic intolerance at the magnitude and duration of head-to-foot acceleration that Artemis astronauts might experience. We will address these knowledge gaps using diuretic-induced hypovolemia as an analog of short-duration spaceflight adaptation in a two-phased approach. First, we will determine whether there are differences between sexes in orthostatic tolerance during 20 minutes of exposure to a range of gravity levels (0.50, 0.75, and 1.00 G) that Artemis astronauts might experience during descent to and ascent from the lunar surface. Due to limited data in the acceleration range from 0.50 to 1.00 G, data from this phase (Study Phase I) will provide evidence to inform NASA standards on the magnitude and duration of sustained accelerations that can be tolerated by all astronauts. Second, we will determine whether lower body compression garments will extend tolerance times and reduce the stress on the cardiovascular system in women and men exposed to the same range of gravity levels and duration of sustained acceleration as used in Study Phase I. These data will further inform NASA standards by increasing the operational range of lunar ascent and descent acceleration limits when the Orion Orthostatic Intolerance Garment (OIG) is used. Thus, this study will determine whether female astronauts will experience a greater risk for developing orthostatic intolerance than male astronauts during descent to or ascent from the lunar surface across a range of gravitational levels and whether the planned compression garment countermeasure can effectively mitigate this risk in all astronauts during Artemis missions. Specific Aims 1) To determine whether there are differences between hypovolemic men and women in orthostatic tolerance (assessed by tilt time) and cardiovascular responses during 20 minutes of exposure to a range of gravity levels (0.50, 0.75, and 1.00 G) simulated by different angles of head-up tilt that Artemis astronauts might experience during descent to and ascent from the lunar surface (Study Phase I). 2) To determine whether the Orion Orthostatic Intolerance Garment (OIG) will protect men and women against orthostatic intolerance (assessed by tilt time) and reduce the cardiovascular responses caused by exposure to the range gravity levels (0.50, 0.75, and 1.00 G) simulated by different angles of head-up tilt that Artemis astronauts might experience during descent to and ascent from the lunar surface (Study Phase II).
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This work is recommended to be performed as a directed study to inform Human Landing System (HLS) requirements in a time efficient manner to meet the schedule for the assessment of proposals provided by the HLS commercial partners. Results from this study will inform the sustained acceleration limits for HLS, including the requirements related to the use of the standing posture during descent to and ascent from the lunar surface and the use of lower body compression garments, e.g., the Orion Orthostatic Intolerance Garment (OIG), to protect crew during exposure to head-to-foot accelerations after exposure to weightlessness.
Research Impact/Earth Benefits:	Results from this study will inform the sustained acceleration limits for HLS, including the requirements related to the use of the standing posture during descent to and ascent from the lunar surface and the use of lower body compression garments, the Orion OIG, to protect against crew during exposure to head-to-foot accelerations after exposure to weightlessness.
Task Progress:	In this reporting year, several important milestones were achieved. First, the protocol was reviewed and approved by the NASA Johnson Space Center (JSC) Institutional Review Board (IRB) and a Test Readiness Review (TRR) was conducted and accepted. Second, the Cardiovascular and Vision Laboratory (CVL) completed preparations needed to initiate the study. This study will assess tilt tolerance and cardiovascular responses in normovolemic and hypovolemic subjects at 30°, 50°, and 80° head-up tilt to simulate gravitational loads that astronauts might experience during descent to and ascent from the lunar surface. Subsequently, we will assess the effectiveness of the NASA Orthostatic Intolerance Garment (OIG) at mitigating cardiovascular stress and/or signs of presyncope at the same tilt angles. The protocol for the carbon monoxide rebreathing (CORB) technique, which has been used to measure red cell mass and plasma volume in our laboratory for many years, was updated to current best practices to improve precision and accuracy. Third, the CVL team initiated data collection, and five subjects will have completed the first phase of the study at the time of this report (October 31, 2022). Phase I includes subject recruitment and familiarization (informed consent), acquiring anthropometric measurements for the construction of the OIG, and tilt testing in the normovolemia condition (control) without the OIG. On a separate day, 20 mg Furosemide is administered intravenously to induce diuresis and fluid volume loss, and tilt tests are performed at three tilt angles simulating <1-G. Phase II activities include a "fit check" of the OIG individually constructed for each subject, and in a separate visit tilt testing while hypovolemic and wearing the OIG at the same tilt angles simulating <1-G as in Phase I. In the coming year, additional male and female subjects will be recruited to assess for differences in orthostatic tolerance at varying tilt angles between sexes (Phase I) and to evaluate the efficacy of lower body compression (Phase II).

Bibliography Type:	Description: (Last Updated: 11/18/2025)
Abstracts for Journals and Proceedings	Lee SMC, Lytle J, Ferguson C, Laurie SS, Young M, Macias. "Effect of sex on orthostatic intolerance and cardiovascular responses during simulated lunar descent and ascent." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023